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Lilac Molecular Laboratory
Cytogenetics

SNP Microarray

Specimen Number:	**659-987-3617-5**	Account Number:	**54924550**
Patient Name:	**LINDQVIST, ROSA**	Ordering Physician:	**SAMUEL OKORO, MD**
Date of Birth:	**03/01/2022**	Specimen Type:	**Extracted DNA from outside laboratory**
Gender:	**Female**	Date Collected:	**12/29/2024**
Patient ID:	**CL48-95153**	Date Received:	**01/01/2025**
Lab Number:	**YU76-35763 G**	Date Reported:	**01/23/2025**
Indications:	**Developmental delay and seizures**		

Test: **Chromosomal Microarray, Pediatric**

Genotyping Targets: 3357702 Array Type: SNP

MICROARRAY RESULT: 5.43 MB DUPLICATION OF 19Q13.12Q13.2

INTERPRETATION: PATHOGENIC

arr[hg19]19q13.12q13.2(37,810,279-43,242,840)x3

Whole-genome SNP microarray analysis found a duplication of the segment shown above, which contains 516 genes.

The change is a gain of genetic material, a duplication, of a large region at 19q13.12q13.2 on chromosome 19. The region contains about five hundred genes, so the duplication changes the amount of several genes at once. Duplications of regions like this one are classified as pathogenic when the region is known to cause developmental problems when it is duplicated. The laboratory classified this change as pathogenic. It is the primary finding of the test.

The chromosomal microarray found an extra copy of a region on chromosome 19, which is called a duplication. The region at 19q13.12q13.2 is large and contains many genes. The laboratory classified the duplication as pathogenic, which means it is considered the cause of the features that led to testing.

Because the region contains many genes, the duplication can affect development in more than one way. Testing of the parents can show whether the duplication was inherited from a parent or arose new in the child. The result should be reviewed with a clinical geneticist, who can plan further care.

Under the current reporting criteria, the array showed no other copy number changes and no copy-neutral regions of homozygosity. **Genetic counseling is recommended.**

For follow-up studies, send the parents' blood samples under test code 160810, labeled with the proband's name, birth date and specimen number. This sample will be kept for 12 months to serve as a positive control in the parental studies.

Methodology: SNP microarray analysis was performed with a whole-genome array of copy number and SNP probes. Genomic DNA was digested, ligated to adaptors, amplified, labeled and hybridized to the array. Data were analyzed against the hg19 (GRCh37) human reference genome. The array detects copy number changes of about 50 kb and larger in targeted regions. It does not detect balanced rearrangements, point mutations or low-level mosaicism.

Additional Information

A pathogenic copy-number change explains the reported features. The change is a duplication, meaning an extra copy, of a region at 19q13.12q13.2 that contains about five hundred genes.

The test looks for chromosome changes that are linked to developmental delay, intellectual disability, seizures and other neurodevelopmental conditions. These conditions differ widely. Features can include delayed speech, delayed motor skills, and differences in growth or in the shape of the face. Copy-number changes happen when a section of a chromosome is lost or present in extra copies. The extra genetic material changes the amount of protein made by the genes in the region, which can disturb brain development.

Test both parents for the same duplication, because the result shows whether the change was inherited or arose new in the child.

Refer the child to a clinical geneticist to review the result with the features that led to testing and to plan further care.

Ask the child's neurologist to review seizure management in light of this result, if the child has seizures.

Information for the Family

The test found an extra copy of a section of chromosome 19. This extra section is the likely reason for the features that led to testing.

The section contains many genes. Extra copies of this kind can affect development in several ways, and the effect can differ from child to child.

Your child's doctor or a genetics specialist can explain what the result means for your child. They can also offer testing to the parents to find out whether the change was inherited.

Electronically signed by Renee Castillo, MD, Director, Cytogenetics, on 01/23/2025.

Provided Clinical Phenotype

As written by the ordering clinician on the test requisition form.

- Seizure
- Muscle hypotonia
- Mental deficiency